

The Blue GPS can be enabled with all cellular phones.

So people will be able to see where their friends are—and have a better idea of estimated arrival time—without having to call them. And, because the wearer doesn't have to be talking on the phone while driving, it will improve safety.

In addition to the button linking the shoe and cell phone users, another button can be pressed for an 'E911' call. When that button is pushed, a signal is sent to IDG's monitoring division, ID Conex, which then attempts to contact the wearer. If the shoe wearer does not respond, ID Conex alerts the proper authorities.

Write in the air with your mobile phone

You can now write short notes in the air with your phone and have that message automatically sent to an e-mail address. This has been made possible by a new cell phone technology called PhonePoint Pen developed by engineering students at Duke University, Durham, USA. To create this application, the students took advantage of the accelerometers in emerging cell phones.

Accelerometers are the devices in phones that not only keep track of the phone's movements but also make it possible for the display screens to rotate from landscape to portrait



PhonePoint Pen



Microsoft's Project Natal demo

modes depending on how the phone is rotated. These devices are always 'on,' so there is no additional burden on the phone to use this new application.

The researchers developed an application that uses the built-in accelerometers in cell phones to recognise human writing. By holding the phone like a pen, you can write short messages or draw simple diagrams in the air. The accelerometer converts the gestures into images, which can be sent to any e-mail address for future reference.

The potential uses are practically limitless. Say, you're in a class and there is an interesting slide on the screen. The researchers foresee being able to take a photo of the slide and write a quick note on it for future reference.

The PhonePoint Pen prototype is expected to be available for download within the next few months.

Gesture-based game control

Microsoft has announced a hands-free motion-sensitive controller system for the Xbox 360. Called Project Natal, it is claimed to be the world's first to combine an RGB camera, depth sensor, multi-array microphone and custom processor running proprietary software all in one device. Unlike 2D cameras and controllers, Project Natal tracks your full body movement in 3D,

while responding to commands, directions and even a change of emotion in your voice.

In addition, the Project Natal sensor is not light-dependent. It can recognise you just by looking at your face, and it doesn't just react to key words but understands what you're saying. Call a play in a football game, and players will actually respond. Want to log onto Xbox LIVE? Simply step in front of the sensor.

World's thinnest MEMS accelerometer

STMicroelectronics has introduced what it claims to be the world's thinnest three-axis digital MEMS accelerometer. The LIS302DLH is a 16-bit device that is 0.75mm thick and shares the 3×5mm² footprint of other devices in ST's Piccolo MEMS family for use in ultra-space-saving product designs. Micro-mechanical components are built on a single chip using robust processes similar to those for conventional semiconductor fabrication. So accelerometers and other devices such as gyroscopes can be produced within extremely small dimensions, and also benefit from low friction, low wear and high shock resistance.

Applications for this accelerometer include motion sensing, orientation awareness, free-fall detection and